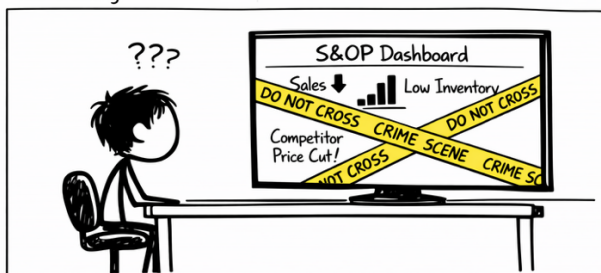


Sales & Operations Planning

A SIMULATION GAME

This tutorial explains how to play the S&OP game

Alex stared at the monthly S&OP dashboard. Sales were down, inventory was tight, and the competitor had cut price again. The numbers weren't just numbers; they were evidence. Somewhere in the system, flow was breaking.



Alex called his team into the cramped conference room. "We're not losing because our product is worse," he said. "We're losing because our system can't keep up with our own demand."



Finance lead pointed to the chart. "When we raised price, our share slipped. When we stocked out, our share slipped. Then the competitor undercut us."

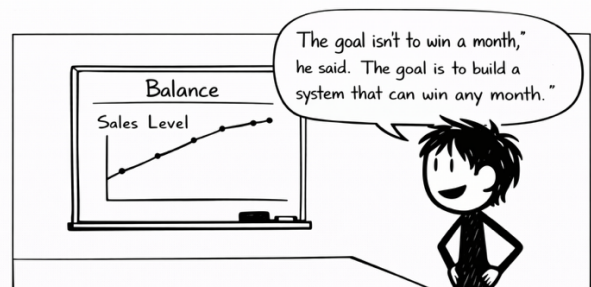
Silence.



They began to see it: every stockout was a ripple. Customers drifted. Share declined. The competitor smelled blood and tightened their pricing. And the system remembered.

What if we stopped thinking in isolated decisions?

Price, availability, and momentum are not separate. They're one system. If we want to win, we have to protect flow.



By month six, the company was back to balance. Not because they found a magic price, but because they learned the first principle: protect flow, then exploit the leverage.

Alex looked at the board and smiled. "The goal isn't to win a month," he said. "The goal is to build a system that can win any month."

1 Understanding the User Interface

The interface is divided into:

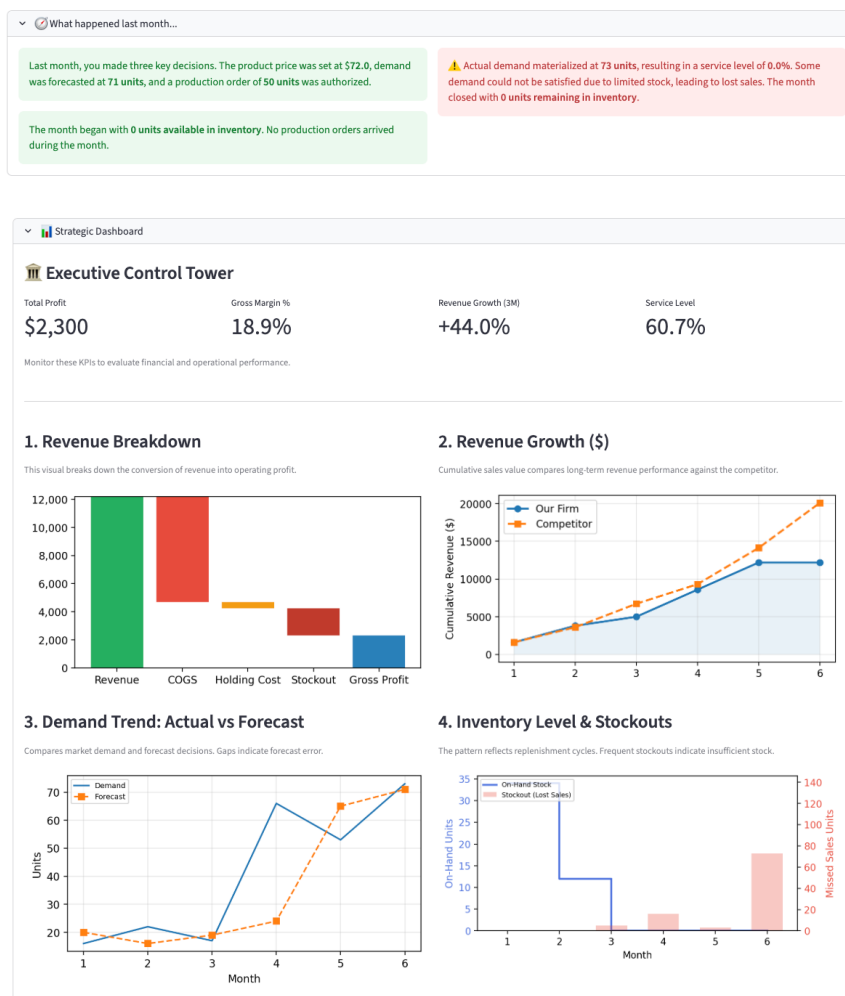
- A feedback panel that explains what happened last month.
- A dashboard panel that summarizes performance over time.
- Decision panels for each role.

You should think of the UI as a management cockpit rather than a form to fill out. Every number shown is there to support a decision. The game is organized as a monthly decision cycle. Each month follows the same sequence:

- Review results from previous months.
- Make sales pricing decisions.
- Make demand forecasts.
- Make production and inventory decisions.
- Execute the decisions and move to the next month.

The UI enforces this order intentionally. Some decisions are locked until others are completed. This reflects real-world dependencies between functions.

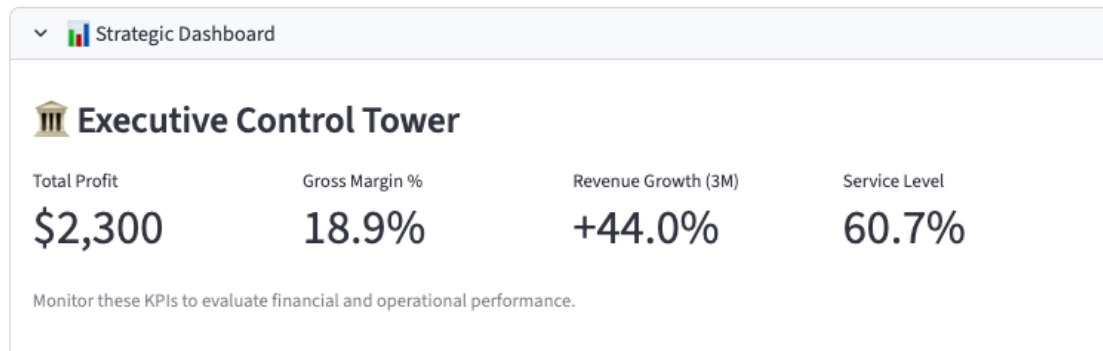
Team: Budhi | Month 7



2 Reading Metrics and Charts

The dashboard is an executive-level view. It answers one question: *Is the system performing well over time?*

2.1 Executive Metrics



The top row shows cumulative performance.

- **Total Profit:** Revenue minus production cost, holding cost, and stockout penalties. This is the main score of the game.
- **Gross Margin %:** Profit as a percentage of revenue. This reflects pricing quality and cost control.
- **Forecast Accuracy:**
- **Inventory Turnover:**
- **Service Level:** Units sold divided by total demand. This measures how well operations support sales.

A strong team grows revenue while keeping service level high and inventory costs controlled.

2.2 Revenue Breakdown

1. Revenue Breakdown

This visual breaks down the conversion of revenue into operating profit.



This waterfall-style chart explains *where profit comes from*.

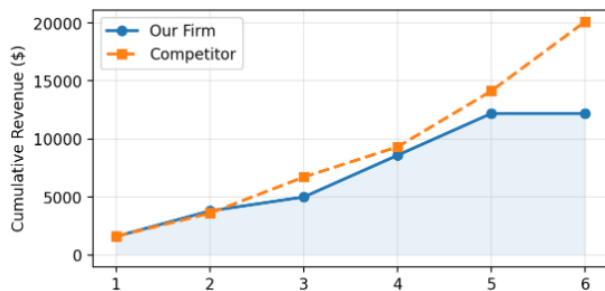
- Revenue is reduced by cost of goods sold.
- Holding cost penalizes excess inventory.
- Stockout penalties penalize missed demand.

If profit is low, this chart tells you whether the problem is pricing, overstocking, or stockouts.

2.3 Revenue Growth

2. Revenue Growth (\$)

Cumulative sales value compares long-term revenue performance against the competitor.



This chart compares cumulative revenue against the competitor.

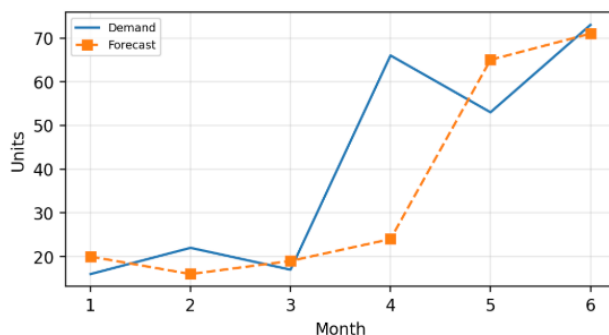
- A widening gap in your favor means your strategy is working.
- A flat or declining gap means either pricing or service level is weak.

It is especially relevant for the sales role, but operations indirectly influence it through service level.

2.4 Demand: Actual vs Forecast

3. Demand Trend: Actual vs Forecast

Compares market demand and forecast decisions. Gaps indicate forecast error.



This chart compares:

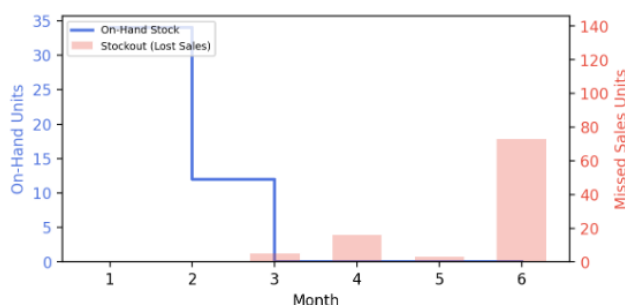
- Actual demand from the market.
- Forecast values entered by the team.

Large and persistent gaps indicate forecasting bias. Forecast accuracy matters because inventory decisions depend directly on it.

2.5 Inventory and Stockouts

4. Inventory Level & Stockouts

The pattern reflects replenishment cycles. Frequent stockouts indicate insufficient stock.



This chart shows a classic inventory sawtooth pattern.

- The step line is on-hand inventory.
- Bars represent lost sales due to stockouts.

Frequent stockouts indicate under-ordering or delayed reactions to demand changes. Very high inventory without stockouts indicates over-ordering.

3 Understanding the Simulation Logic

The engine translates decisions into outcomes. It combines market behavior with supply chain constraints.

3.1 Market Demand

Total market demand evolves over time using a growth curve and seasonal effects. Demand is not constant, and it cannot be fully controlled. Your firm only captures a **share** of this market. Market share is driven by three factors:

- Relative price versus the competitor.
- Popularity (market share), which adds inertia.
- Availability (service level), which rewards reliable fulfillment.

Even a low price cannot fully compensate for repeated stockouts. Likewise, perfect service cannot overcome extreme overpricing.

3.2 Competitor Behavior

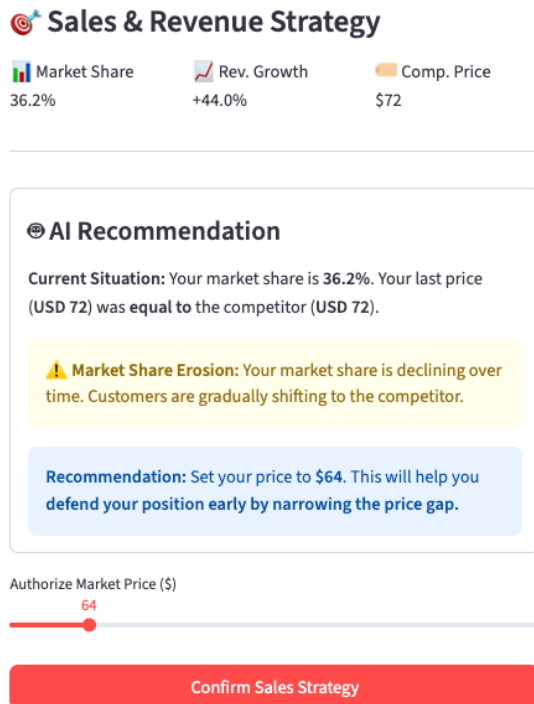
The competitor adapts based on recent market share.

- When market share is balanced, the competitor will attempt to mimic your price.
- When losing share, the competitor discounts aggressively.
- When winning strongly, the competitor raises prices to harvest profit.

This means your decisions shape the competitor's behavior indirectly.

4 S&OP Planners

4.1 Sales Planner



Primary responsibility: Set the market price to balance revenue growth and profitability. The Sales Manager controls only one decision: **price**. Despite its simplicity, price has system-wide effects.

Key KPIs for Sales

- **Market Share (3-month average).** This shows how attractive your offering is relative to the competitor. It reacts slowly because customer behavior has inertia.
- **Revenue Growth (3-month average).** This captures short-term momentum. Declining growth signals pricing or service problems.
- **Competitor Price (last month).** This is a reference, not a target. Large gaps invite aggressive competitor responses.

How Sales Logic Works

- Lower prices increase demand, but only if product availability is reliable.
- Repeated stockouts reduce service level, which directly lowers market share in future months.
- Higher prices improve margin per unit but reduce demand.

How to Read the AI Recommendation

The AI evaluates (i) Recent market share level, (ii) Direction of market share change, (iii) Revenue growth trend. Based on this, it classifies the situation into recovery, yield, erosion, or stability modes. The recommended price is a *reference anchor*, not a command.

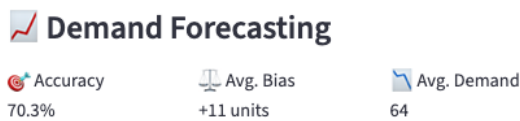
Common Sales Mistakes

- Chasing market share with deep discounts while operations cannot support volume.
- Reacting to one bad month instead of observing three-month patterns.
- Ignoring competitor behavior feedback loops.

Good Sales Practices

- Make incremental price changes.
- Coordinate price moves with operations capacity.
- Accept short-term volume loss if it stabilizes profit and forecasts.

4.2 Demand Planner



Primary responsibility: Translate sales strategy into a realistic demand signal. The forecast is the backbone of the system. All inventory decisions depend on it.

AI Recommendation

Current Situation: Baseline demand is 64 units. At a price of \$64.0, demand is projected to increase by 10 units. Additionally, the rising trend adds +3 units.

Warning: Low forecast accuracy (70.3%). Consider reviewing demand volatility.

Recommendation: Enter 78 units. This reflects trend, price impacts, and bias correction.

Authorize Demand Forecast (Units)

78

Lock Forecast

Key KPIs for Forecasting

- **Forecast Accuracy (3-month average):** Measures how close forecasts are to realized demand.
- **Average Bias (3-month average):** Indicates systematic over- or under-forecasting.
- **Demand Baseline (3-month average):** Provides scale and context for adjustments.

How to Read the AI Recommendation.

The AI forecast combines four elements (i) Recent demand level, (ii) Price impact from the sales decision, (iii) Bias correction based on recent errors, (iv) Short-term demand trend. Forecasts are intentionally conservative. Extreme jumps are dampened unless evidence is consistent. Warnings appear when accuracy drops below a safe threshold or a consistent bias is detected.

Common Forecasting Mistakes

- Simply copying last month's demand.
- Overreacting to one-time spikes.
- Ignoring the effect of price changes.

Good Forecasting Practices

- Focus on bias reduction before chasing accuracy.
- Ask Sales to explain pricing decision, not just the number.
- Prefer smooth adjustments unless a structural change is clear.

4.3 Supply Planner

 **Supply & Replenishment**

 **Fill Rate**
100.0%

 **On-Hand Stock**
161

 **Next Arrival**
50

AI Recommendation

Current Situation: You have 161 units on hand. Your total Pipeline (On-hand + all in-transit) is 261 units, providing 3.0 months of total forecast coverage.

 **Balanced Supply:** Your pipeline is healthy and aligned with forecast trends.

Recommendation: Authorize 100 units to reach 2 months of coverage.

Production Order (Arrival: Month 10)

100 - +

Authorize Production

Primary responsibility: Ensure product availability at minimum total cost. Operations controls **production order quantity**, but must think two months ahead.

Key KPIs for Operations

- **Fill Rate (last month):** Percentage of demand fulfilled. This affects future market share.
- **On-Hand Stock:** What is immediately available to sell.
- **Pipeline Inventor:** On-hand plus all in-transit orders. This is the true inventory position.
- **Months of Coverage:** Pipeline divided by forecast demand.

How Inventory Logic Works

- Production has a fixed two-month lead time. Orders placed today arrive 2 months later.
- Inventory available in a month equals last month's ending inventory plus any arriving orders.
- Sales are limited by available inventory; unmet demand results in stockouts.
- Stockouts reduce service level and incur penalties, while excess inventory incurs holding costs. Both reduce profit.

How to Read the AI Recommendation

The AI evaluates: (i) Current on-hand stock (ii) Arrivals next month (iii) Total pipeline coverage relative to forecast. It targets roughly 2.5 months of coverage *after* lead-time demand.

Common Operations Mistakes

- Ordering based only on on-hand inventory.
- Ignoring demand during lead time.
- Overcorrecting after a single stockout.

Good Operations Practices

- Think in pipelines, not on-hand stock.
- Trust forecasts unless bias is proven.
- Absorb small forecast errors with inventory buffers.

5 Parameters and Rules

5.1 Parameters

Category	Parameter	Value	Description	Implications
Supply Chain	Production lead time	2 months	Time between placing an order and receiving inventory	Forces forward-looking inventory planning
	Initial inventory	50 units	Starting on-hand stock at Month 1	Buffers early demand uncertainty
	Pipeline inventory	On-hand + in-transit	Total inventory position	Correct basis for inventory decisions
	Target coverage	2 months	Desired pipeline coverage	Balances service and cost
Inventory Cost	Unit production cost	50 per unit	Cost incurred for each unit sold	Sets minimum viable price
	Holding cost	10 per unit per month	Cost of carrying unsold inventory	Penalizes over-ordering
	Stockout penalty	20 per unit	Penalty for each unit of unmet demand	Penalizes under-ordering more strongly
Market Demand	Base market demand	1000 units	Long-term market capacity	Caps total demand
	Market growth curve	Logistic (S-curve)	Market grows over time	Encourages early positioning
	Seasonality factor	Quarterly multipliers	Demand fluctuates by quarter	Requires adaptive forecasting
Market Competition	Market share weight	$\beta = 1.0$	Influence of past market share	Adds inertia to customer behavior
	Service level weight	$\gamma = 0.5$	Impact of fulfillment reliability	Rewards operational excellence
	Price sensitivity	$\alpha = 2.0$	Weight of relative price in choice	Drives competitive pricing dynamics
Forecasting	Baseline Demand	Holt's method	History used for bias and trend	Smooths short-term noise
	Trend detection	Holt's method	Detects short-term demand drift	Supports adaptive forecasts
	Bias correction	3 months window	Adjusts consistent errors	Prevents systematic mis-planning

5.2 Formula/Decision Rules

Area	Output Variable	Formula (Conceptual)	Key Parameters	Implications
Market Growth	Total market demand	Logistic growth × seasonal factor	Base market demand, growth rate, seasonality	Market expands over time but with limits
Choice Model	Market share	$1 / (1 + e^{-(\text{utility difference})})$	α, β, γ	Customers choose probabilistically
Firm Demand	Firm demand	Total market × firm share	Market demand, share	Sales potential before inventory limits
Inventory Availability	Available stock	Beginning inventory + arrivals	Lead time	Supply constraint
Actual Sales	Units sold	$\min(\text{Available stock, firm demand})$	Inventory, demand	Sales limited by inventory
Stockout Volume	Missed demand	$\max(0, \text{demand} - \text{units sold})$	Demand, inventory	Lost sales
Pipeline Inventory	Pipeline units	Ending inventory + in-transit	Lead time	True inventory position
Forecast Error	Error	Actual demand – forecast	Forecast	Measures bias
Months of Coverage	MOC	Pipeline inventory / forecast	Forecast	Inventory adequacy indicator
Revenue	Revenue	Units sold × price	Price	Top-line performance
Cost of Goods Sold	COGS	Units sold × unit cost	Unit cost	Variable production cost
Holding Cost	Holding cost	Ending inventory × holding cost	Holding cost per unit	Penalty for excess stock
Stockout Penalty	Penalty	Missed demand × penalty cost	Stockout penalty	Penalty for poor service
Profit	Operating profit	Revenue – (COGS + holding + penalty)	All cost parameters	Overall performance